
MUON+: Towards More Effective Muon via One Additional Normalization Step for LLM Pre-training

Ruijie Zhang, Yequan Zhao, Ziyue Liu, Zhengyang Wang,
Yupeng Su, Liyan Tan, Zheng Zhang[†]
University of California at Santa Barbara
ruijiezh@ucsb.edu,
zhengzhang@ece.ucsb.edu

Abstract

Muon has recently emerged as a strong optimizer for large language model pre-training, orthogonalizing the momentum matrix via Newton–Schulz polar iterations. A natural intuition is that polar iterations, by flattening the singular spectrum to all ones, should also eliminate column- and row-wise norm imbalance in the update. We show that this is not true in practice: practical polar steps can substantially *amplify* the imbalance. We term this the *post-polar imbalanced update* problem, and prove that such imbalance tightens the second-order term in a blockwise descent analysis, weakening Muon’s per-step descent guarantee. Motivated by this analysis, we propose MUON+, a one-line fix that inserts a single normalization step after polar orthogonalization. MUON+ adds no optimizer state. Across pre-training experiments on GPT and LLaMA models from 60M to 7B parameters, spanning both compute-optimal budgets and extended token-to-parameter ratios up to approximately 200, MUON+ consistently outperforms Muon in terms of training and validation perplexity, leading to significant overall pre-training speedup.

1 Introduction

Based on the empirical observation of scaling laws [16, 13, 20], powerful foundation models such as GPT, DeepSeek, LLaMA, and Gemini [1, 26, 9, 38] have been trained and widely deployed. Nevertheless, as the sizes of both model parameters and training datasets reach extreme levels, the computational cost of pre-training has become prohibitively high. This challenge has motivated increasing research dedicated to improving pre-training efficiency [31, 11, 6, 49, 46, 29], with a particular emphasis on the critical role of optimizers. Although Adam [18] and AdamW [30] are still the dominant optimizers, numerous efficient optimizers have been proposed to reduce the computing or memory cost of large-scale pre-training [18, 30, 27, 15, 43, 42, 22, 23, 34, 21, 24, 33].

Among these, the Muon optimizer [15] has recently emerged as a strong choice for pre-training. Its key idea is to orthogonalize the momentum matrix via Newton–Schulz iterations, flattening its singular spectrum to mitigate “rank collapse” of the update. Muon has been shown to scale to massive foundation models [28] and is now integral to production-scale systems such as Kimi and GLM [39, 8, 44, 40]. A rapidly growing literature further studies Muon’s efficiency, scalability, and theoretical properties [47, 3, 17, 2, 25, 19].

Muon update rule. Unlike Adam/SGD-based optimizers, Muon [15] operates on a matrix rather than a vector. By enforcing orthogonalization on the gradient, Muon prevents the rank collapse of the gradient by replacing the singular value matrix with an identity matrix. Let η and μ denote the learning rate and the momentum coefficient, respectively. Assume that $\mathbf{W}_t \in \mathbb{R}^{m \times n}$ is the layer being adapted at iteration t , $\mathbf{G}_t \in \mathbb{R}^{m \times n}$ is its stochastic gradient, and $\mathbf{M}_t$ is the gradient momentum

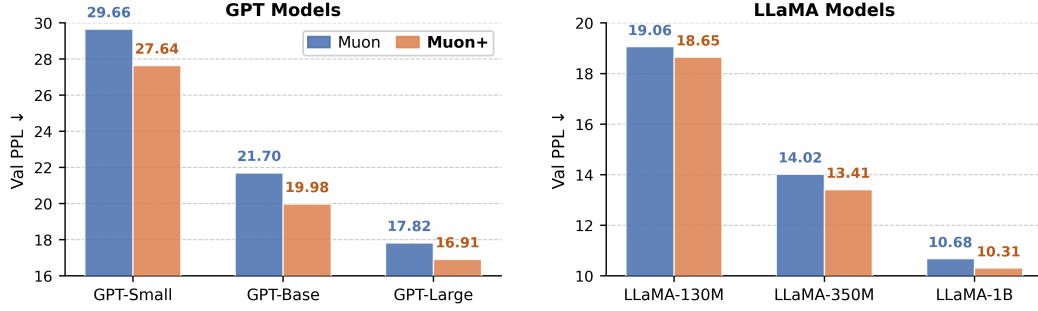


Figure 1: Pre-training GPT and LLaMA models at scales ranging from 130M to 1B parameters under compute-optimal settings. Quantitative results are provided in Section 3. **MUON+** consistently outperforms MUON across all runs. We also conduct larger-scale and overtraining experiments for both GPT and LLaMA; the results are presented in Section 3.

at iteration t . The Muon update is given by

$$\begin{aligned}
 \mathbf{M}_t &= \mu \mathbf{M}_{t-1} + (1 - \mu) \mathbf{G}_t \\
 \mathbf{O}_t &= \text{Ortho}(\mathbf{M}_t) \\
 \mathbf{W}_t &= \mathbf{W}_{t-1} - \eta \cdot \sqrt{m/n} \cdot \mathbf{O}_t
 \end{aligned} \tag{1}$$

where $\text{Ortho}(\cdot)$ denotes the semi-orthogonal matrix function closest to the input matrix [12]. Specifically, if the SVD of the input matrix $\mathbf{M}$ is $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, then $\text{Ortho}(\mathbf{M}) := \mathbf{U}\mathbf{V}^T$. In practice, the Newton-Schulz iteration process [12] is commonly used to approximate the SVD. The dimensional pre-factor $\sqrt{m/n}$ was suggested by [3] for better scalability.

In this work, we identify a limitation of Muon, which we term the *post-polar imbalanced update* problem, and propose a simple yet effective extension, **MUON+**, to address it. We summarize our contributions as follows:

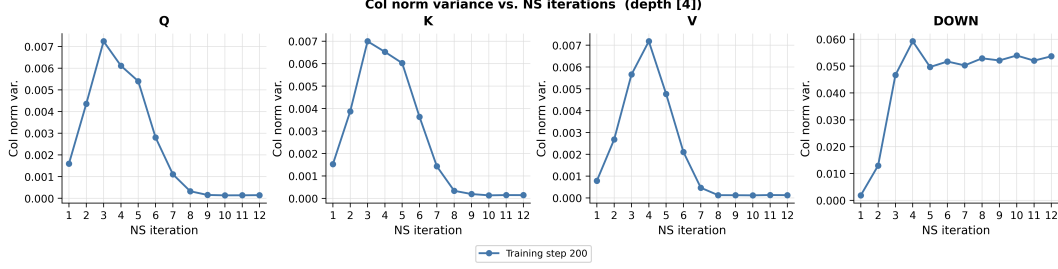
- **A counter-intuitive phenomenon in Muon: update imbalance.** We identify the *post-polar imbalanced update* problem: under practical Newton-Schulz settings, polar iterations can *amplify* rather than reduce column/row norm imbalance.
- **Theoretical analysis.** We prove (i) that update imbalance weakens Muon’s convergence (Theorem 2.3), and (ii) that the expected column/row variance shift of one polar step admits a closed-form characterization with a provably positive amplification region (Lemma 2.1 and Corollary 2.2).
- **A simple yet effective fix: MUON+.** A single post-orthogonalization normalization mitigates the imbalance with zero memory overhead and negligible computing overhead.
- **Empirical validation on GPT and LLaMA.** On GPT and LLaMA pre-training from 60M to 7B parameters, spanning compute-optimal budgets and token-to-parameter ratios up to ~ 200 , **MUON+** consistently outperforms Muon across all evaluated settings. It reduces validation perplexity by up to **2.02**, improves average downstream zero-shot accuracy by up to **1.3%** across **7** tasks, and speeds up the pre-training up to **37.1%**, while requiring **zero** additional optimizer states.

2 Method

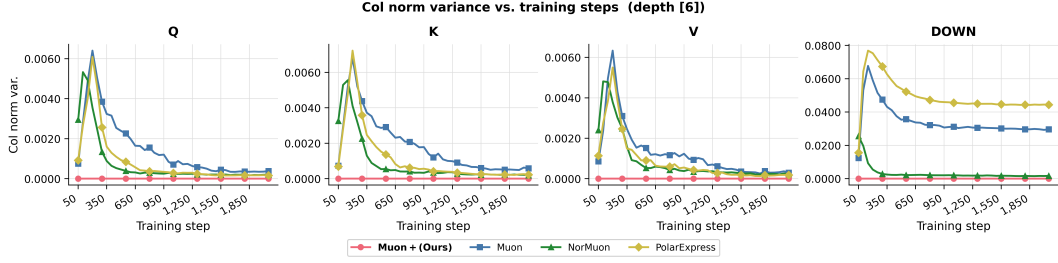
This section first shows that Muon’s polar iteration can *amplify* column/row update imbalance (Section 2.1), then proves that such imbalance weakens Muon’s local descent guarantees (Section 2.2), and finally presents **MUON+** as a fix (Section 2.3). Proofs are deferred to Appendices A and B.

2.1 Update Imbalance Caused by Muon’s Polar Iterations

Update imbalance. Muon is designed to reduce *spectral* imbalance by flattening the singular values of the update matrix. In this work, we study a complementary notion of imbalance in the parameter space. We define *update imbalance* as highly non-uniform magnitudes across the elements of the



(a) Variance trend with respect to the number of Newton–Schulz iterations. The imbalance remains amplified even under a typical number of Newton–Schulz iterations (e.g., 5). For non-square layers, the variance in one direction cannot be reduced to zero.



(b) Variance trend under different Muon variants. Existing variants fail to resolve the issue, whereas Muon+ maintains zero variance throughout training (see Figure 12 for more details).

Figure 2: Comparison of variance behavior under different settings during the realistic training of a 60M LLaMA model on 1.1B FineWeb tokens.

update matrix (e.g., some columns or rows having much larger ℓ_2 norm than others (Fig. 3)). We measure this imbalance by the variance of column or row norms. For any matrix $\mathbf{X}$, define

$$s(\mathbf{X}) := (\|\mathbf{X}_{1:}\|_2^2, \dots, \|\mathbf{X}_{n:}\|_2^2).$$

The row-norm imbalance is measured by

$$\text{Var}(s(\mathbf{X})) = \frac{1}{n} \sum_{i=1}^n \left(\|\mathbf{X}_{i:}\|_2^2 - \frac{1}{n} \|\mathbf{X}\|_F^2 \right)^2.$$

The column-norm imbalance can be defined likewise.

Post-polar imbalanced update. For a square layer, Muon replaces $\mathbf{M} = \mathbf{U}\Sigma\mathbf{V}^\top$ with $\mathbf{Q} = \mathbf{U}\mathbf{V}^\top$, whose rows and columns all have unit ℓ_2 norm. Ideally, one would expect that the polar step should eliminate row/column norm imbalance in the update $\mathbf{Q}$. In practice we find the opposite: with a typical number of Newton-Schulz iterations (5 steps), the variance is significantly amplified (Figure 2). In principle, one may mitigate the imbalance by running a large number of Newton-Schulz iterations, but this is computationally infeasible in LLM pre-training. For a non-squared matrix, we cannot reduce the variance of one direction to 0 even if we use exact SVD.

To understand why this phenomenon occurs, we characterize the variance shift induced by

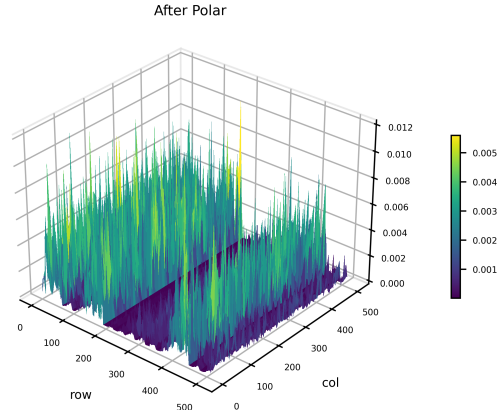


Figure 3: Layer-6 Q Projection Update matrix of Muon. The update matrix is highly unbalanced. The trends over iteration/training are shown in Figure 2.

one polar step. Let

$$\mathbf{M} \in \mathbb{R}^{n \times n}, \quad \|\mathbf{M}\|_F^2 = 1, \quad \mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

where

$$\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_n), \quad 1 \geq \sigma_k \geq 0,$$

Define one Newton-Schulz step by

$$\mathbf{Q} := a\mathbf{M} + b(\mathbf{M}\mathbf{M}^\top)\mathbf{M} + c(\mathbf{M}\mathbf{M}^\top)^2\mathbf{M},$$

with coefficients $a, b, c \in \mathbb{R}$. Introduce the quintic polynomial

$$\varphi(x) := ax + bx^3 + cx^5.$$

Then

$$\mathbf{Q} = \mathbf{U}\varphi(\mathbf{\Sigma})\mathbf{V}^\top.$$

Lemma 2.1 (Expected variance shift of one polar step). *Assume that $\mathbf{U}$ is Haar distributed on the orthogonal group¹ Let $\mathbf{M}$ and $\mathbf{Q}$ be defined as above, and let*

$$\lambda_k := \sigma_k^2, \quad g(\lambda) := \varphi(\sqrt{\lambda})^2 = \lambda(a + b\lambda + c\lambda^2)^2.$$

Then, for each pair $1 \leq i < j \leq n$, there exists

$$\xi_{ij} \in (\min\{\lambda_i, \lambda_j\}, \max\{\lambda_i, \lambda_j\})$$

such that

$$\mathbb{E}[\text{Var}(s(\mathbf{Q})) - \text{Var}(s(\mathbf{M}))] = \frac{2}{n^2(n+2)} \sum_{1 \leq i < j \leq n} (\lambda_i - \lambda_j)^2 (g'(\xi_{ij})^2 - 1). \quad (2)$$

Corollary 2.2 (Existence of a positive expected variance shift). *Let $\varphi(x)$ and $g(\lambda)$ be defined as above. Assume that $a = \varphi'(0) > 1$. Then there exists $\delta > 0$ such that, whenever*

$$\lambda_1, \dots, \lambda_n \in [0, \delta]$$

are not all equal,

$$\mathbb{E}[\text{Var}(s(\mathbf{Q})) - \text{Var}(s(\mathbf{M}))] > 0.$$

In particular, the expected variance shift of one polar step can be positive.

Detailed proof is provided in Appendix B

Remark. Lemma 2.1 and Corollary 2.2 suggest a natural partition of the spectral interval into amplification and reduction regions. Specifically, the derivative profile of g , which is fully determined by the Newton-Schulz coefficients a, b, c , divides the spectrum into intervals where $|g'(\lambda)| > 1$ and intervals where $|g'(\lambda)| < 1$. This partition determines the sign of the pairwise contributions to the variance shift: pairs associated with the former regions contribute positively, whereas pairs associated with the latter regions contribute negatively. As an example, consider the commonly used coefficients

$$(a, b, c) = (2, -1.5, 0.5).$$

Figure 4 plots $g'(\lambda)$ together with the thresholds ± 1 . By Lemma 2.1, each eigenvalue pair (λ_i, λ_j) contributes through an intermediate point ξ_{ij} , with sign determined by $g'(\xi_{ij})^2 - 1$. Thus, pairs with $|g'(\xi_{ij})| > 1$ increase the variance, while those with $|g'(\xi_{ij})| < 1$ decrease it. The overall variance shift is the sum of these pairwise effects. As the number of Newton-Schulz iterations grows, all λ_k converge to 1, eventually entering the right variance-reduction region in Figure 4. Therefore, the variance decreases asymptotically as the number of iterations tends to infinity.

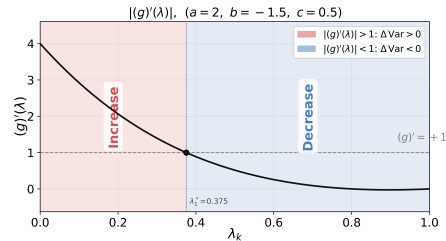
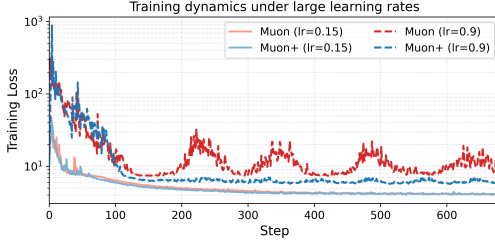
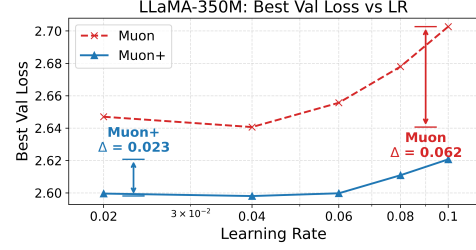


Figure 4: Regions of imbalance amplification and reduction. When ξ_{ij} lies in the red (blue) region, the imbalance is amplified (reduced). We also provide figures for other commonly used coefficients in Appendix D.

¹The orthogonal group is the group of all real $n \times n$ matrices $\mathbf{U}$ satisfying $\mathbf{U}^\top \mathbf{U} = \mathbf{I}$.



(a) Training behaviors under different learning rates. Muon becomes unstable as the learning rate increases, whereas MUON+ remains stable.



(b) Validation loss of LLaMA 350M. The loss of Muon increases rapidly as the learning rate increases, whereas MUON+ remains more stable.

Figure 5: Stable step-size comparison between Muon and MUON+. MUON+ remains stable under larger learning rates, while Muon exhibits unstable training and degraded validation loss.

We also investigate the variance trend with respect to the number of Newton–Schulz iterations under real training dynamics in Figure 2a. As shown in the figure, when the number of Newton–Schulz iterations is insufficient, the variance can be significantly amplified; in particular, for a commonly used choice of 5 iterations, the variance still remains amplified. More importantly, even in the limit of infinitely many iterations, for a non-square layer, the variance along one direction cannot be reduced to zero (e.g., the down-projection layer in Figure 2a). We will later show that Muon+ can further improve performance exactly in this regime: applying Muon+ along the direction whose variance cannot be eliminated still brings additional gains (see Table 8).

We further examine variants such as NorMuon[25] and PolarExpress[2] under the same realistic pre-training dynamics. Figure 2b shows that these variants still fail to eliminate the variance imbalance.

2.2 Imbalanced Updates Hurt Convergence in Muon

An imbalanced update can make optimization less effective. Intuitively, excessively large updates may cause instability, while excessively small updates leave some parts of the network nearly unchanged. We formalize this intuition using a *blockwise descent analysis* [41].

Theorem 2.3 (Imbalance weakens Muon local descent guarantee). *Fix any partition of the mn entries of $\mathbf{W}_t$ into B disjoint blocks, and for any vectorization $\mathbf{x} \in \mathbb{R}^{mn}$ of matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$, let $[\mathbf{x}]_b$ denote its elements in block b . Consider the Muon-style orthogonalized update*

$$\mathbf{W}_{t+1} = \mathbf{W}_t - \eta_t \mathbf{O}_t, \quad \mathbf{O}_t := \text{Ortho}(\nabla L(\mathbf{W}_t)).$$

Under local smoothness and a local near-block-diagonal Hessian structure with respect to the chosen partition, the one-step loss decrease satisfies

$$L(\mathbf{W}_{t+1}) - L(\mathbf{W}_t) \leq -\eta_t \|\nabla L(\mathbf{W}_t)\|_* + \frac{\eta_t^2}{2} \tilde{\Lambda}_{O,t} + \mathcal{O}(\eta_t^3 \|\mathbf{O}_t\|_F^3),$$

where $\|\cdot\|_*$ denotes the nuclear norm, and $\tilde{\Lambda}_{O,t}$ is the *block-weighted Hessian curvature* at step t :

$$\tilde{\Lambda}_{O,t} := \sum_{b=1}^B \left\| [\nabla^2 L(\mathbf{W}_t)]_b \right\|_2 \left\| [\mathbf{o}_t]_b \right\|_2^2, \quad \mathbf{o}_t := \text{vec}(\mathbf{O}_t), \quad (3)$$

Prior work [48, 14, 45, 7, 41] has identified a positive *gradient–Hessian correlation*: parameter blocks with larger gradient norms tend to have larger diagonal Hessian blocks. We further find that this correlation is preserved after the polar step in Muon (see Appendix A.5). Under this correlation, larger imbalance in the column gradient norms leads to a larger $\tilde{\Lambda}_{O,t}$ and hence a smaller largest stable step size. A precise definition and the full proof are provided in Appendix A.

The above theoretical analysis is consistent with our experimental observation in Figure 5: update imbalance reduces the largest stable step size of Muon, whereas our proposed MUON+ (which will be explained soon) reduces this imbalance and thus improves training stability and performance.

2.3 An Easy-to-Deploy Solution: MUON+

The above theoretical analysis and experimental demonstration shows that variance imbalance can hurt the performance of Muon. To address this issue, we propose MUON+, a simple yet highly effective modification by adding a single normalization step after Polar iterations.

MUON+ follows the Muon update rule in Eq. (1), while additionally applying a normalization step to the orthogonalized update in order to reduce the imbalance in the update matrix:

$$\begin{aligned}\mathbf{M}_t &= \mu \mathbf{M}_{t-1} + (1 - \mu) \mathbf{G}_t, \\ \mathbf{O}_t &= \text{Norm}_{(d)}(\text{Ortho}(\mathbf{M}_t)), \\ \mathbf{W}_t &= \mathbf{W}_{t-1} - \eta \cdot \sqrt{m/n} \cdot \mathbf{O}_t.\end{aligned}\tag{4}$$

Here $\text{Norm}_{(d)}(\cdot)$ denotes a normalization operator applied along direction d . We provide the pseudocode as in Algorithm 1 in the Appendix. We consider column-wise and row-wise normalization, denoted by $\text{Norm}_{(\text{col})}(\cdot)$ and $\text{Norm}_{(\text{row})}(\cdot)$, respectively. For $\mathbf{X} = [x_{ij}] \in \mathbb{R}^{m \times n}$, define

$$\text{Norm}_{(\text{col})}(\mathbf{X}) := \mathbf{X} \mathbf{D}_{\text{col}}^{-1}, \quad \mathbf{D}_{\text{col}} := \text{diag} \left(\sqrt{\sum_{i=1}^m x_{i1}^2}, \dots, \sqrt{\sum_{i=1}^m x_{in}^2} \right),$$

and

$$\text{Norm}_{(\text{row})}(\mathbf{X}) := \mathbf{D}_{\text{row}}^{-1} \mathbf{X}, \quad \mathbf{D}_{\text{row}} := \text{diag} \left(\sqrt{\sum_{j=1}^n x_{1j}^2}, \dots, \sqrt{\sum_{j=1}^n x_{mj}^2} \right).$$

We further consider composed normalizations, such as $\text{Norm}_{(\text{col_row})}$ and $\text{Norm}_{(\text{row_col})}$, defined by sequential application of the two normalizations.

3 Experiments

We evaluate MUON+ on two widely adopted architectures: GPT and LLaMA. Our evaluation covers both compute-optimal pre-training and long-horizon overtraining regimes, followed by systematic ablation studies in Section 3.4. Note that, all the experiments in this paper use 5 iterations in $\text{Ortho}(\cdot)$ to approximate $\mathbf{U}\mathbf{V}^T$.

3.1 Pre-training in Compute-Optimal Settings

GPT Models. We first evaluate MUON+ on GPT-style models. We pre-train GPT-Small, GPT-Base, and GPT-Large with a compute-optimal [13] token-to-parameter (T2P) ratio. All models are trained on the FineWeb dataset [32], tokenized using the GPT tokenizer, with a vocabulary size of 50,257 and a batch size of 512. Training is conducted on H100/A100 GPUs using mixed precision (bfloat16). Following the setup in [2], we apply MUON+ (or Muon) to all parameters except embeddings, unembeddings, normalization layers, and positional encodings, which are optimized using AdamW. For the polar operator, we adopt the same configuration as in [15]. We sweep normalization directions under learning rates in $[0.003, 0.005, 0.01, 0.02, 0.04]$ for both MUON+ and Muon except GPT-Huge, and report the best results in Table 1. GPT-Huge is trained with a learning rate of 0.01 using bfloat16 precision (no mixed precision). Detailed hyperparameters and full sweep results are provided in Appendix E and Appendix F. As shown in Table 1, MUON+ consistently outperforms Muon across all GPT model scales.

Model	Param (M)	Train Tokens (B)	Muon	MUON+
GPT-Small	124	3.0	29.66	27.64 (-2.02)
GPT-Base	362	7.2	21.70	19.98 (-1.72)
GPT-Large	774	15.5	17.82	16.91 (-0.91)
GPT-Huge (bf16)	6,654	20.0	15.90	14.69 (-1.21)

Table 1: Validation PPL of Muon vs. MUON+ on GPT models.

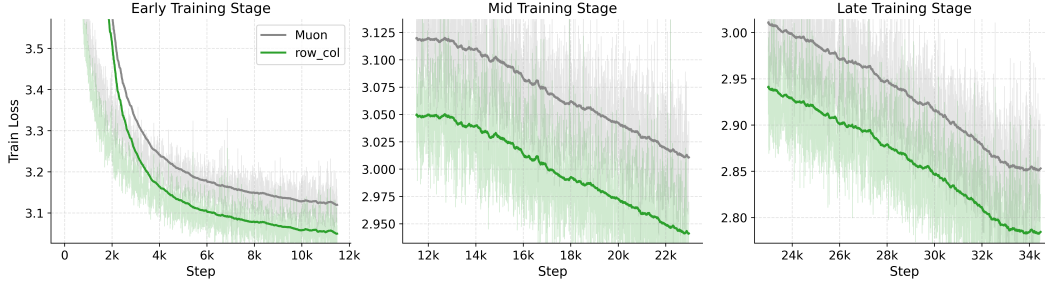


Figure 6: Training loss curves under overtraining for GPT-Base.

LLaMA Models. To extend our evaluation beyond GPT architectures, we benchmark our proposed approach against AdamW and Muon by pre-training LLaMA-based language models. This validation is also conducted on the FineWeb dataset, spanning model capacities from 60M up to 7B parameters (architectural details are provided in Table 11).

Based on the compute-optimal scaling guidelines established by [13], we strictly pair model sizes with training token budgets: the 60M, 130M, 350M, 1B, and 7B parameter models are trained on 1.1B, 2.2B, 6.4B, 13.1B, and 19.8B tokens, respectively. Across all configurations, we maintain a constant batch size of 512 and employ the LLaMA-2 tokenizer with a 32,000-token vocabulary. In line with the setup described in Section 3.1, all experiments are executed using mixed precision on H100 and A100 GPUs.

We sweep normalization directions over learning rates in $[0.005, 0.01, 0.02, 0.04, 0.06, 0.08]$ for all models in this section (Section 3.4), except LLaMA-1B and LLaMA-7B due to computational constraints. For LLaMA-1B, we sweep only the *col-row* and *row-col* variants. For LLaMA-7B, we use bfloat16 precision and the *row-col* variant with a learning rate of 0.01. The results are reported in Table 2. Additional hyperparameter details are provided in Appendix E. Overall, as shown in Table 2, MUON+ consistently outperforms the baselines, achieving the best overall performance across all evaluated scales. We also report mean $\pm$ standard deviation over 5 random seeds for the GPT-Base and LLaMA-350M models to assess the statistical robustness of MUON+ improvements (see Table 18).

Model	Param (M)	Train Tokens (B)	AdamW	Muon	MUON+
LLaMA-60M	58	1.1	33.10	25.75	25.25 (-0.50)
LLaMA-130M	134	2.2	23.64	19.06	18.65 (-0.41)
LLaMA-350M	368	6.4	16.18	14.02	13.41 (-0.61)
LLaMA-1B	1339	13.1	14.38	10.68	10.31 (-0.37)
LLaMA-7B (bf16)	6,738	19.8	–	10.54	10.12 (-0.42)

Table 2: Validation PPL of Muon vs. MUON+ on LLaMA models.

3.2 Pre-Training with High Token-to-Parameter Ratios

Model	Param (M)	Train Tokens (B)	Muon	MUON+
GPT-Base	362	72	16.97	15.84 (-1.13)
LLaMA-350M	368	72	11.48	11.03 (-0.45)

Table 3: Overtraining GPT-Base/LLaMA-350M. Models are trained on 72 billion FineWeb tokens.

We further study MUON+ in the overtraining regime using GPT-Base and LLaMA-350M. Both models are trained on 72 billion FineWeb tokens, corresponding to a token-to-parameter ratio of approximately 200. Detailed hyperparameters are provided in Appendix E.

Table 3 shows that MUON+ outperforms Muon for both models in the overtraining regime. The improvement remains at a token-to-parameter ratio of approximately 200, indicating that the additional normalization continues to be effective under extended training. Figure 6 shows that the gap is stable throughout training.

	Ave.	OBQA	HellaSwag	ARC-E	WSC	Winogrande	BoolQ	PIQA
Muon	0.481	0.306	0.446	0.444	0.375	0.522	0.574	0.700
MUON+	0.494	0.320	0.480	0.471	0.365	0.534	0.577	0.712

Table 4: Downstream evaluation results on GPT-Base overtraining checkpoint.

Downstream Evaluations. Furthermore, we compare MUON+ with Muon on downstream tasks using the pretrained models. We evaluate their commonsense reasoning performance, with results reported in Table 4. MUON+ outperforms Muon on most tasks and achieves higher average accuracy.

3.3 Training Efficiency

MUON+ has nearly the same per-step runtime and memory cost as Muon (see Appendix 20). In this section, we report the wall-clock time required to reach the same target loss, where the target is set to the final training loss of the Muon baseline. We use the same data scale as in Table 1 and Table 2. The results are shown in Table 5. MUON+ speeds up pre-training by up to 37.1% in the evaluated settings.

Model	Target loss	Muon steps	Muon+ steps	Speed-up
LLaMA-130M	3.060	4228	3448	↑22.6%
LLaMA-350M	2.788	3064	2374	↑29.1%
GPT-Base	3.144	3447	2515	↑37.1%
GPT-Large	2.927	3710	3032	↑22.4%

Table 5: Training cost required to reach a target loss.

3.4 Ablation Study

To better understand the source of performance gains, we conduct systematic ablations on the key design choices of MUON+. In particular, we analyze the effects normalization directions and orthogonalization methods while keeping all other training settings fixed.

Model	None	Col	Row	Col-Row	Row-Col
LLaMA-60M	25.75	25.34	25.29	25.25	25.25
LLaMA-130M	19.34	19.16	18.98	18.65	18.68
LLaMA-350M	14.02	13.73	13.46	13.41	13.44

Table 6: Best validation perplexity under different normalization directions. For each model, we report the best result across learning rates. Lower is better.

Impact of Normalization Directions. We study the impact of different normalization directions, including none(Muon Baseline), col, row, col_row, and row_col, across multiple model scales, see Appendix F for more quantitative results. As shown in Table 6, introducing normalization leads to better optimization behavior compared to the baseline Muon across all evaluated settings. This improvement maintains as the model scale increases. Importantly, applying bi-directional normalization consistently outperforms single-directional normalization.

Ablation for Polar Methods Ortho($\cdot$). To validate the robustness of MUON+ under different polar functions, we adopt 3 different methods (all with 5 iterations) in this section: You [6], Jordan[15] and the more recent PolarExpress[2]. Hyperparameters are identical as in Table 14. As shown in Table 7, MUON+ consistently outperforms Muon across all evaluated methods. As discussed in Section 2.1, one direction may remain unbalanced for a non-square matrix even when $\mathbf{UV}^T$ is computed exactly via SVD. Therefore, applying MUON+ on top of this exact orthogonal update can still yield additional gains. This is experimentally shown in Table 8.

Model	Ortho($\cdot$)	Muon	MUON+
LLaMA-350M	You	14.01	13.38(-0.63)
	Jordan	14.02	13.41(-0.61)
	PolarExpress	13.90	13.27(-0.63)

Table 7: Validation perplexity comparison among Muon, and MUON+(col_row) on LLaMA-350M; all methods use 5 iterations.

Model	Ortho($\cdot$)	Muon	MUON+
GPT-Small	SVD	29.17	27.90(-1.27)

Table 8: MUON+ and Muon with exact SVD. Learning rate is 0.005 for all runs. MUON+ uses col_row as its normalization direction.

4 Related Work and Comparison with MUON+

Muon [15] has shown promising performance in large-scale pre-training. Follow-up work can be grouped into three directions.

- **Optimizing the polar function (complementary to Muon+).** Several methods improve Muon by designing better polar iterations. You [6] learns per-iteration coefficients, POLAREXPRESS [2] derives minimax-optimal Newton–Schulz coefficients, and TURBO-MUON [5] manipulates the spectrum for faster convergence of the polar iteration. Because Muon+ does not change anything inside the polar functions, these methods are complementary to MUON+. As shown in Table 7, combining MUON+ with these methods always yields additional performance improvements.
- **Pre-conditioning techniques (potential competitors with Muon+).** Some recent techniques improve Muon with additional preconditioning. NorMuon [25] adds a per-neuron second moment, AdaMuon [35] combines Muon with an Adam-style element-wise step, and Mano [10] projects updates onto the oblique manifold. These methods are closer competitors to MUON+, as they also aim to improve the update geometry. As shown in Table 9, MUON+ consistently outperforms these competing baselines.². Notably, unlike NorMuon and AdaMuon, MUON+ does not require additional optimizer states.
- **Theoretical foundations of Muon (relevant but not competing with Muon+).** Existing theoretical studies mainly explain Muon from spectral or non-Euclidean optimization views. [4] frames Muon as steepest descent under the spectral norm; [19] analyzes Muon through a non-Euclidean trust-region formulation; [37] models Muon using an isotropic curvature approximation. These results do not capture the properties of the parameter-space. We provide an alternative perspective: this work explains why and how Muon can amplify the parameter-space imbalance (Lemma 2.1, Corollary 2.2) and why such imbalance provably hurts convergence (Theorem 2.3). Our analysis also helps explain why preconditioning methods like NorMuon and AdaMuon can be effective, as their preconditioning mechanisms implicitly reduce the parameter-space imbalance induced by Muon (see Figure 2b).

Method	GPT-Small	GPT-Base
Muon+	27.64	19.98
Turbo-Muon	29.69	21.91
NorMuon	28.44	21.31
AdaMuon	29.27	22.38

Table 9: Validation PPL ($\downarrow$) across different Muon methods.

5 Conclusion

In this work, we have shown that polar iterations can amplify the imbalance in the *parameter space* of the update in Muon. Through a blockwise descent analysis, we have shown that such an imbalance reduces Muon’s largest stable step size. To address this issue, we have proposed MUON+, a simple yet effective modification that adds a single normalization step after orthogonalization. Across GPT and LLaMA pre-training from 60M to 7B parameters, MUON+ has consistently improved pre-training performance with zero memory overhead and significant overall training speedup.

²For all experiments in this comparison except AdaMuon, we swept the learning rate over [0.005, 0.01, 0.02, 0.04, 0.06, 0.08]. For AdaMuon, which requires smaller learning rates, we swept [0.005, 0.003, 0.001]. We report the best result for each method.

References

- [1] J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, S. Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [2] N. Amsel, D. Persson, C. Musco, and R. M. Gower. The polar express: Optimal matrix sign methods and their application to the muon algorithm. *arXiv preprint arXiv:2505.16932*, 2025.
- [3] J. Bernstein. Deriving muon, 2025.
- [4] J. Bernstein and L. Newhouse. Modular duality in deep learning. *arXiv preprint arXiv:2410.21265*, 2024.
- [5] T. Boissin, T. Massena, F. Mamalet, and M. Serrurier. Turbo-muon: Accelerating orthogonality-based optimization with pre-conditioning. *arXiv preprint arXiv:2512.04632*, 2025.
- [6] F. L. Cesista, J. You, and K. Jordan. Squeezing 1–2% efficiency gains out of muon by optimizing the newton-schulz coefficients, 2025. URL <http://leloykun.github.io/ponder/muon-opt-coeffs/>.
- [7] M. Crawshaw, M. Liu, F. Orabona, W. Zhang, and Z. Zhuang. Robustness to unbounded smoothness of generalized signsgd. *Advances in neural information processing systems*, 35:9955–9968, 2022.
- [8] D. Ding, Z. Ju, Y. Leng, S. Liu, T. Liu, Z. Shang, K. Shen, W. Song, X. Tan, H. Tang, et al. Kimi-audio technical report. *arXiv preprint arXiv:2504.18425*, 2025.
- [9] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Vaughan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- [10] Y. Gu and Z. Xie. Mano: Restriking manifold optimization for llm training. *arXiv preprint arXiv:2601.23000*, 2026.
- [11] A. Han, J. Li, W. Huang, M. Hong, A. Takeda, P. K. Jawanpuria, and B. Mishra. Sltrain: a sparse plus low rank approach for parameter and memory efficient pretraining. *Advances in Neural Information Processing Systems*, 37:118267–118295, 2024.
- [12] N. J. Higham. *Functions of matrices: theory and computation*. SIAM, 2008.
- [13] J. Hoffmann, S. Borgeaud, A. Mensch, E. Buchatskaya, T. Cai, E. Rutherford, D. de Las Casas, L. A. Hendricks, J. Welbl, A. Clark, et al. Training compute-optimal large language models. In *Proceedings of the 36th International Conference on Neural Information Processing Systems*, pages 30016–30030, 2022.
- [14] K. Jiang, D. Malik, and Y. Li. How does adaptive optimization impact local neural network geometry? *Advances in Neural Information Processing Systems*, 36:8305–8384, 2023.
- [15] K. Jordan, Y. Jin, V. Boza, J. You, F. Cesista, L. Newhouse, and J. Bernstein. Muon: An optimizer for hidden layers in neural networks, 2024.
- [16] J. Kaplan, S. McCandlish, T. Henighan, T. B. Brown, B. Chess, R. Child, S. Gray, A. Radford, J. Wu, and D. Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.
- [17] A. Khaled, K. Ozkara, T. Yu, M. Hong, and Y. Park. Muonbp: Faster muon via block-periodic orthogonalization. *arXiv preprint arXiv:2510.16981*, 2025.
- [18] D. P. Kingma. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- [19] D. Kovalev. Understanding gradient orthogonalization for deep learning via non-euclidean trust-region optimization. *arXiv preprint arXiv:2503.12645*, 2025.

- [20] T. Kumar, Z. Ankner, B. F. Spector, B. Bordelon, N. Muennighoff, M. Paul, C. Pehlevan, C. Re, and A. Raghunathan. Scaling laws for precision. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [21] X. Li. Black box lie group preconditioners for sgd, 2022.
- [22] X.-L. Li. Preconditioned stochastic gradient descent. *IEEE Transactions on Neural Networks and Learning Systems*, 29(5):1454–1466, May 2018.
- [23] X.-L. Li. Preconditioner on matrix lie group for sgd, 2018.
- [24] X.-L. Li. Stochastic hessian fittings with lie groups, 2024.
- [25] Z. Li, L. Liu, C. Liang, W. Chen, and T. Zhao. Normuon: Making muon more efficient and scalable. *arXiv preprint arXiv:2510.05491*, 2025.
- [26] A. Liu, B. Feng, B. Xue, B. Wang, B. Wu, C. Lu, C. Zhao, C. Deng, C. Zhang, C. Ruan, et al. Deepseek-v3 technical report. *arXiv preprint arXiv:2412.19437*, 2024.
- [27] H. Liu, Z. Li, D. L. W. Hall, P. Liang, and T. Ma. Sophia: A scalable stochastic second-order optimizer for language model pre-training. In *The Twelfth International Conference on Learning Representations*, 2024.
- [28] J. Liu, J. Su, X. Yao, Z. Jiang, G. Lai, Y. Du, Y. Qin, W. Xu, E. Lu, J. Yan, et al. Muon is scalable for llm training. *arXiv preprint arXiv:2502.16982*, 2025.
- [29] Z. Liu, R. Zhang, Z. Wang, M. Yan, Z. Yang, P. D. Hovland, B. Nicolae, F. Cappello, S. Tang, and Z. Zhang. Cola: Compute-efficient pre-training of llms via low-rank activation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 4627–4645, 2025.
- [30] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.
- [31] F. Mehmood, S. Ahmad, and T. K. Whangbo. An efficient optimization technique for training deep neural networks. *Mathematics*, 11(6):1360, 2023.
- [32] G. Penedo, H. Kydlíček, A. Lozhkov, M. Mitchell, C. A. Raffel, L. Von Werra, T. Wolf, et al. The fineweb datasets: Decanting the web for the finest text data at scale. *Advances in Neural Information Processing Systems*, 37:30811–30849, 2024.
- [33] T. Pethick, W. Xie, K. Antonakopoulos, Z. Zhu, A. Silveti-Falls, and V. Cevher. Training deep learning models with norm-constrained lmos, 2025.
- [34] O. Pooladzandi and X.-L. Li. Curvature-informed sgd via general purpose lie-group preconditioners, 2024.
- [35] C. Si, D. Zhang, and W. Shen. Adamuon: Adaptive muon optimizer. *arXiv preprint arXiv:2507.11005*, 2025.
- [36] C. Spearman. The proof and measurement of association between two things. 1961.
- [37] W. Su. Isotropic curvature model for understanding deep learning optimization: Is gradient orthogonalization optimal? *arXiv preprint arXiv:2511.00674*, 2025.
- [38] G. Team, R. Anil, S. Borgeaud, J.-B. Alayrac, J. Yu, R. Soricut, J. Schalkwyk, A. M. Dai, A. Hauth, K. Millican, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.
- [39] K. Team, Y. Bai, Y. Bao, G. Chen, J. Chen, N. Chen, R. Chen, Y. Chen, Y. Chen, Y. Chen, et al. Kimi k2: Open agentic intelligence. *arXiv preprint arXiv:2507.20534*, 2025.
- [40] K. Team, A. Du, B. Yin, B. Xing, B. Qu, B. Wang, C. Chen, C. Zhang, C. Du, C. Wei, et al. Kimi-vl technical report. *arXiv preprint arXiv:2504.07491*, 2025.

- [41] A. Tomihari and I. Sato. Understanding why adam outperforms sgd: Gradient heterogeneity in transformers. *arXiv e-prints*, pages arXiv–2502, 2025.
- [42] N. Vyas, D. Morwani, R. Zhao, I. Shapira, D. Brandfonbrener, L. Janson, and S. M. Kakade. SOAP: Improving and stabilizing shampoo using adam. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [43] H. Yuan, Y. Liu, S. Wu, X. Zhou, and Q. Gu. Mars: Unleashing the power of variance reduction for training large models, 2024.
- [44] A. Zeng, X. Lv, Q. Zheng, Z. Hou, B. Chen, C. Xie, C. Wang, D. Yin, H. Zeng, J. Zhang, et al. Glm-4.5: Agentic, reasoning, and coding (arc) foundation models. *arXiv preprint arXiv:2508.06471*, 2025.
- [45] J. Zhang, T. He, S. Sra, and A. Jadbabaie. Why gradient clipping accelerates training: A theoretical justification for adaptivity. *arXiv preprint arXiv:1905.11881*, 2019.
- [46] R. Zhang, Z. Liu, Z. Wang, and Z. Zhang. Lax: Boosting low-rank training of foundation models via latent crossing. *arXiv preprint arXiv:2505.21732*, 2025.
- [47] R. Zhang, Y. Zhao, Z. Liu, Z. Wang, D. Li, Y. Su, S. Liu, and Z. Zhang. Teon: Tensorized orthonormalization beyond layer-wise muon for large language model pre-training. *arXiv preprint arXiv:2601.23261*, 2026.
- [48] Y. Zhang, C. Chen, T. Ding, Z. Li, R. Sun, and Z.-Q. Luo. Why transformers need adam: A hessian perspective. *Advances in neural information processing systems*, 37:131786–131823, 2024.
- [49] J. Zhao, Z. Zhang, B. Chen, Z. Wang, A. Anandkumar, and Y. Tian. Galore: Memory-efficient llm training by gradient low-rank projection. *arXiv preprint arXiv:2403.03507*, 2024.

A Blockwise Descent Analysis for Muon

This appendix follows the blockwise descent framework of [41]. To align with the notation of prior work, we identify the matrix parameter $\mathbf{W} \in \mathbb{R}^{m \times n}$ with a vectorization

$$\theta := \text{vec}(\mathbf{W}) \in \mathbb{R}^P, \quad P = mn, \quad \|\theta\|_2 = \|\mathbf{W}\|_F.$$

Throughout this appendix, we fix an arbitrary partition of the index set $\{1, \dots, P\}$ into B non-overlapping blocks

$$\{1, \dots, P\} = I_1 \sqcup I_2 \sqcup \dots \sqcup I_B, \quad P_b := |I_b|.$$

For any $x \in \mathbb{R}^P$ we write $[x]_b := x|_{I_b} \in \mathbb{R}^{P_b}$, so $\|x\|_2^2 = \sum_{b=1}^B \|[x]_b\|_2^2$. Two canonical choices (the ones used in Section 2.1) are:

- *Column blocks*: $B = n$, $P_b = m$, and $[\theta]_b = \mathbf{W}_{:,b}$ (i.e., θ is the column-stacking vectorization and each block is a column of $\mathbf{W}$);
- *Row blocks*: $B = m$, $P_b = n$, and $[\theta]_b = \mathbf{W}_{b,:}^\top$ (i.e., θ is the row-stacking vectorization and each block is a row of $\mathbf{W}$).

A.1 Block-Hessian and weighted curvature

For each $b \in \{1, \dots, B\}$, let $[\nabla^2 L(\theta)]_b \in \mathbb{R}^{P_b \times P_b}$ denote the b -th principal submatrix of the Hessian under the chosen partition, and define the block-diagonal part of the Hessian by

$$\nabla^2 L_D(\theta) := \text{blockdiag}(\{[\nabla^2 L(\theta)]_b\}_{b=1}^B).$$

For the Muon-style orthogonalized update

$$\mathbf{O}_t := \text{Ortho}(\nabla L(\mathbf{W}_t)), \quad o_t := \text{vec}(\mathbf{O}_t) \in \mathbb{R}^P,$$

we define the update-weighted block curvature at iterate θ_t by

$$\tilde{\Lambda}_{O,t} := \sum_{b=1}^B \left\| [\nabla^2 L(\theta_t)]_b \right\|_2 \|[o_t]_b\|_2^2. \quad (5)$$

Since $\mathbf{O}_t$ is semi-orthogonal (if $\nabla L(\mathbf{W}_t) = \mathbf{U}\Sigma\mathbf{V}^\top$, then $\mathbf{O}_t = \mathbf{U}\mathbf{V}^\top$), we have

$$\|o_t\|_2^2 = \|\mathbf{O}_t\|_F^2 = \min(m, n) =: r.$$

A.2 Local assumptions

We use the following local analogues of Assumptions 4.2 and 4.3 of [41].

Assumption A.1 (Local Hessian Lipschitz). There exist a neighborhood $\mathcal{N}_t \subseteq \mathbb{R}^P$ of θ_t , containing the line segment

$$\{\theta_t + s(\theta_{t+1} - \theta_t) : s \in [0, 1]\},$$

and a constant $\rho_H > 0$ such that

$$\|\nabla^2 L(\theta) - \nabla^2 L(\theta')\|_2 \leq \rho_H \|\theta - \theta'\|_2 \quad \text{for all } \theta, \theta' \in \mathcal{N}_t.$$

Assumption A.2 (Local near block-diagonal Hessian). With respect to the chosen block partition $\{I_b\}_{b=1}^B$, there exists $\delta_D \geq 0$ such that, for all $\theta \in \mathcal{N}_t$,

$$\|\nabla^2 L(\theta) - \nabla^2 L_D(\theta)\|_2 \leq \delta_D.$$

A.3 Technical lemma

Lemma A.3. Under Assumption A.1, for any $\theta, \theta' \in \mathcal{N}_t$,

$$L(\theta') \leq L(\theta) + \nabla L(\theta)^\top (\theta' - \theta) + \frac{1}{2} (\theta' - \theta)^\top \nabla^2 L(\theta) (\theta' - \theta) + \frac{\rho_H}{6} \|\theta' - \theta\|_2^3. \quad (6)$$

Proof. Define

$$\nu(s) := \theta + s(\theta' - \theta), \quad s \in [0, 1].$$

By the fundamental theorem of calculus,

$$L(\theta') - L(\theta) = \int_0^1 \nabla L(\nu(s))^\top (\theta' - \theta) ds.$$

Also,

$$\nabla L(\nu(s)) - \nabla L(\theta) = \int_0^s \nabla^2 L(\nu(r))(\theta' - \theta) dr.$$

Substituting this identity yields

$$L(\theta') - L(\theta) = \nabla L(\theta)^\top (\theta' - \theta) + \int_0^1 \int_0^s (\theta' - \theta)^\top \nabla^2 L(\nu(r))(\theta' - \theta) dr ds.$$

Add and subtract $\nabla^2 L(\theta)$ inside the quadratic form. The main term becomes

$$\int_0^1 \int_0^s (\theta' - \theta)^\top \nabla^2 L(\theta)(\theta' - \theta) dr ds = \frac{1}{2} (\theta' - \theta)^\top \nabla^2 L(\theta)(\theta' - \theta).$$

For the remainder, Assumption A.1 gives

$$\|\nabla^2 L(\nu(r)) - \nabla^2 L(\theta)\|_2 \leq \rho_H \|\nu(r) - \theta\|_2 = \rho_H r \|\theta' - \theta\|_2.$$

Hence

$$|(\theta' - \theta)^\top (\nabla^2 L(\nu(r)) - \nabla^2 L(\theta))(\theta' - \theta)| \leq \rho_H r \|\theta' - \theta\|_2^3.$$

Integrating over $0 \leq r \leq s \leq 1$ gives

$$\int_0^1 \int_0^s \rho_H r \|\theta' - \theta\|_2^3 dr ds = \frac{\rho_H}{6} \|\theta' - \theta\|_2^3.$$

Combining the terms proves (6). $\square$

A.4 One-step descent bound: formal statement and proof

Theorem A.4 (Theorem 2.3 is restated). *Under Assumptions A.1 and A.2, the Muon-style orthogonalized update*

$$\theta_{t+1} = \theta_t - \eta_t o_t, \quad o_t := \text{vec}(\text{Ortho}(\nabla L(\mathbf{W}_t))), \quad \eta_t > 0,$$

satisfies

$$L(\theta_{t+1}) \leq L(\theta_t) - \eta_t \|\nabla L(\mathbf{W}_t)\|_* + \frac{\eta_t^2}{2} (\tilde{\Lambda}_{O,t} + \delta_D \|o_t\|_2^2) + \frac{\eta_t^3 \rho_H}{6} \|o_t\|_2^3, \quad (7)$$

where $\|\cdot\|_*$ denotes the nuclear norm and $\|o_t\|_2^2 = \|\mathbf{O}_t\|_F^2 = \min(m, n) = r$.

Remark. For simplicity, Theorem 2.3 is stated under the assumption $\delta_D \rightarrow 0$, which does not affect the conclusion.

Proof. Let

$$g_t := \nabla L(\theta_t), \quad \mathbf{O}_t := \text{Ortho}(\nabla L(\mathbf{W}_t)), \quad o_t := \text{vec}(\mathbf{O}_t).$$

Applying Lemma A.3 with $\theta = \theta_t$ and $\theta' = \theta_{t+1} = \theta_t - \eta_t o_t$, we obtain

$$L(\theta_{t+1}) \leq L(\theta_t) - \eta_t g_t^\top o_t + \frac{\eta_t^2}{2} o_t^\top \nabla^2 L(\theta_t) o_t + \frac{\eta_t^3 \rho_H}{6} \|o_t\|_2^3. \quad (8)$$

We handle the first- and second-order terms separately.

Step 1: First-order term. Write the (thin) SVD of $\nabla L(\mathbf{W}_t)$ as

$$\nabla L(\mathbf{W}_t) = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top,$$

so that $\mathbf{O}_t = \mathbf{U}\mathbf{V}^\top$. Since $g_t^\top o_t$ is the Euclidean inner product of the vectorizations of $\nabla L(\mathbf{W}_t)$ and $\mathbf{O}_t$ under a common vectorization convention, it equals the matrix Frobenius inner product, so

$$g_t^\top o_t = \langle \nabla L(\mathbf{W}_t), \mathbf{O}_t \rangle_F = \text{tr}(\mathbf{V}\Sigma^\top \mathbf{U}^\top \mathbf{U}\mathbf{V}^\top) = \text{tr}(\Sigma) = \|\nabla L(\mathbf{W}_t)\|_*,$$

Step 2: Second-order term. By Assumption A.2,

$$\|\nabla^2 L(\theta_t) - \nabla^2 L_D(\theta_t)\|_2 \leq \delta_D.$$

Therefore,

$$o_t^\top \nabla^2 L(\theta_t) o_t = o_t^\top \nabla^2 L_D(\theta_t) o_t + o_t^\top (\nabla^2 L(\theta_t) - \nabla^2 L_D(\theta_t)) o_t,$$

and the second term is bounded by

$$|o_t^\top (\nabla^2 L(\theta_t) - \nabla^2 L_D(\theta_t)) o_t| \leq \delta_D \|o_t\|_2^2.$$

Hence

$$o_t^\top \nabla^2 L(\theta_t) o_t \leq o_t^\top \nabla^2 L_D(\theta_t) o_t + \delta_D \|o_t\|_2^2. \quad (9)$$

Since $\nabla^2 L_D(\theta_t)$ is block-diagonal with respect to the chosen partition, its quadratic form decomposes blockwise:

$$o_t^\top \nabla^2 L_D(\theta_t) o_t = \sum_{b=1}^B [o_t]_b^\top [\nabla^2 L(\theta_t)]_b [o_t]_b.$$

For each block, the operator norm gives

$$[o_t]_b^\top [\nabla^2 L(\theta_t)]_b [o_t]_b \leq \|[\nabla^2 L(\theta_t)]_b\|_2 \| [o_t]_b \|_2^2.$$

Summing over b , we obtain

$$o_t^\top \nabla^2 L_D(\theta_t) o_t \leq \sum_{b=1}^B \|[\nabla^2 L(\theta_t)]_b\|_2 \| [o_t]_b \|_2^2 = \tilde{\Lambda}_{O,t}.$$

Combining this with (9) gives

$$o_t^\top \nabla^2 L(\theta_t) o_t \leq \tilde{\Lambda}_{O,t} + \delta_D \|o_t\|_2^2.$$

Substituting the first-order identity from Step 1 and this bound into (8) proves (7). $\square$

Using $\|o_t\|_2 = \|\mathbf{O}_t\|_F$, Theorem A.4 is equivalently the matrix statement

$$L(\mathbf{W}_{t+1}) \leq L(\mathbf{W}_t) - \eta_t \|\nabla L(\mathbf{W}_t)\|_* + \frac{\eta_t^2}{2} (\tilde{\Lambda}_{O,t} + \delta_D \|\mathbf{O}_t\|_F^2) + \frac{\eta_t^3 \rho_H}{6} \|\mathbf{O}_t\|_F^3,$$

where $\mathbf{O}_t = \text{Ortho}(\nabla L(\mathbf{W}_t))$ and $\|\mathbf{O}_t\|_F^2 = \min(m, n)$. Instantiating the partition $\{I_b\}_{b=1}^B$ as either the columns or the rows of $\mathbf{W}_t$ recovers the column- and row-block versions used in Section 2.1: the principal Hessian blocks $[\nabla^2 L(\theta_t)]_b$ and the absolute squared block norms $\|[o_t]_b\|_2^2$ in (5) then specialize to column- or row-indexed quantities, respectively, while δ_D is the near-block-diagonal slack of the Hessian under the chosen partition.

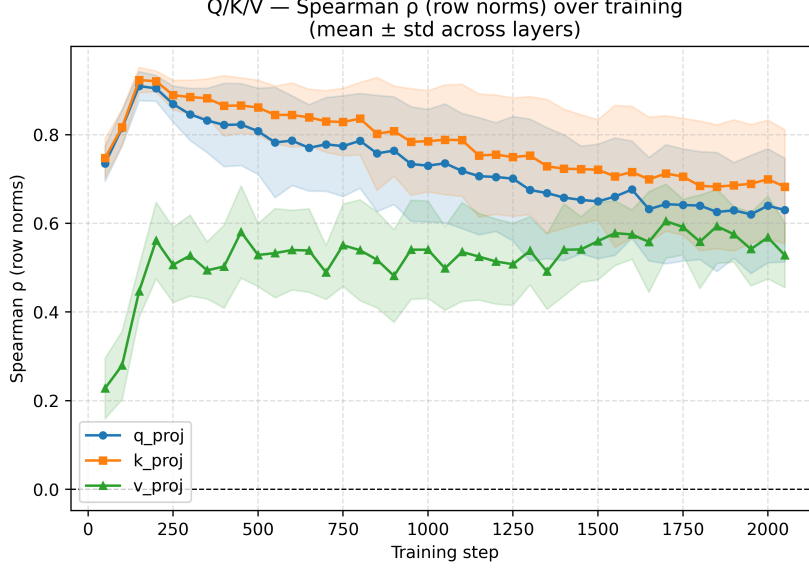


Figure 7: Spearman rank correlation between pre-polar and post-polar (using 5 NS iterations) row norms over training (mean $\pm$ standard deviation across layers). All correlations remain positive throughout training, indicating that rows that are large before the polar step tend to remain large afterward.

A.5 Positive Correlation Before and After the Polar Step

Prior work has identified a positive *gradient–Hessian correlation*: parameter blocks with larger gradient norms tend to be associated with larger diagonal Hessian blocks [48, 14, 45, 7, 41]. Motivated by this observation, we empirically examine whether the polar step preserves the *relative blockwise magnitudes* of the pre-polar update. If it does, then the blocks that are large before the polar step will still tend to remain large afterward.

To quantify this effect, for each matrix updated by Muon, let

$$g_j := \|[\mathbf{G}]_{j,:}\|_2, \quad q_j := \|[\mathbf{Q}]_{j,:}\|_2,$$

denote the row norms of the pre-polar input $\mathbf{G}$ and the post-polar output $\mathbf{Q}$, respectively. We then compute the Spearman rank correlation [36] between the two sequences:

$$\rho_{\text{sp}}(g, q) := \text{Corr}(\text{rank}(g_1, \dots, g_n), \text{rank}(q_1, \dots, q_n)). \quad (10)$$

A positive value of ρ_{sp} indicates that rows that are relatively large before the polar step also tend to remain relatively large afterward; a larger value indicates stronger preservation of the blockwise ordering.

Figure 7 reports the Spearman correlation over training, using row norms and averaging across layers. We observe that the correlations remain consistently positive throughout training. Moreover, the correlations are generally substantial. This shows that the polar step does not destroy the relative blockwise ordering of the pre-polar update.

B Existence of Positive Expected Variance Shift

B.1 Proof of Lemma 2.1

Proof. We divide the proof into four steps. Throughout, let $\lambda_k := \sigma_k^2$ and $g(\lambda) := \varphi(\sqrt{\lambda})^2 = \lambda(a + b\lambda + c\lambda^2)^2$.

Step 1: Row energies in terms of $\mathbf{U}$ and the spectrum.

Since $\mathbf{M} = \mathbf{U}\Sigma\mathbf{V}^\top$ and $\mathbf{Q} = \mathbf{U}\varphi(\Sigma)\mathbf{V}^\top$, the row energies $s_i(\cdot) = \|\cdot\|_2^2$ are

$$s_i(\mathbf{M}) = \sum_{k=1}^n \lambda_k u_{ik}^2, \quad s_i(\mathbf{Q}) = \sum_{k=1}^n g(\lambda_k) u_{ik}^2.$$

Step 2: Expected variance under Haar measure.

We compute $\mathbb{E}[\text{Var}(s(\mathbf{M}))]$ and $\mathbb{E}[\text{Var}(s(\mathbf{Q}))]$ simultaneously; both have the form $s_i = \sum_k \alpha_k u_{ik}^2$ with $\alpha_k = \lambda_k$ or $\alpha_k = g(\lambda_k)$, respectively. Assume $\mathbf{U}$ is Haar-distributed on $O(n)$, each row $(u_{i1}, \dots, u_{in})$ is uniform on S^{n-1} . The relevant spherical moments are

$$\mathbb{E}[u_{ik}^2] = \frac{1}{n}, \quad \mathbb{E}[u_{ik}^4] = \frac{3}{n(n+2)}, \quad \mathbb{E}[u_{ik}^2 u_{i\ell}^2] = \frac{1}{n(n+2)} \quad (k \neq \ell),$$

which follow from exchangeability of coordinates and the constraint $\sum_k u_{ik}^2 = 1$.

The first moment of the row energy is $\mathbb{E}[s_i] = \frac{1}{n} \sum_k \alpha_k$, independent of i . For the second moment,

$$\begin{aligned} \mathbb{E}[s_i^2] &= \sum_k \alpha_k^2 \mathbb{E}[u_{ik}^4] + \sum_{k \neq \ell} \alpha_k \alpha_\ell \mathbb{E}[u_{ik}^2 u_{i\ell}^2] \\ &= \frac{3}{n(n+2)} \sum_k \alpha_k^2 + \frac{1}{n(n+2)} \sum_{k \neq \ell} \alpha_k \alpha_\ell \\ &= \frac{1}{n(n+2)} \left(2 \sum_k \alpha_k^2 + \left(\sum_k \alpha_k \right)^2 \right), \end{aligned}$$

where we used $3 \sum_k \alpha_k^2 + \sum_{k \neq \ell} \alpha_k \alpha_\ell = 2 \sum_k \alpha_k^2 + (\sum_k \alpha_k)^2$. Note that $\mathbb{E}[s_i^2]$ is also independent of i by exchangeability of coordinates under Haar measure. Moreover, the total row energy is deterministic: by orthogonality of $\mathbf{U}$,

$$\sum_{i=1}^n s_i = \sum_{k=1}^n \alpha_k \sum_{i=1}^n u_{ik}^2 = \sum_{k=1}^n \alpha_k,$$

which does not depend on $\mathbf{U}$. Therefore $\bar{s} := \frac{1}{n} \sum_i s_i$ is a deterministic constant, and $\mathbb{E}[\bar{s}^2] = \bar{s}^2 = (\mathbb{E}[s_i])^2$. It follows that

$$\begin{aligned} \mathbb{E}[\text{Var}(s)] &= \mathbb{E}[s_i^2] - (\mathbb{E}[s_i])^2 = \frac{1}{n(n+2)} \left(2 \sum_k \alpha_k^2 + \left(\sum_k \alpha_k \right)^2 \right) - \frac{1}{n^2} \left(\sum_k \alpha_k \right)^2 \\ &= \frac{2}{n^2(n+2)} \left(n \sum_k \alpha_k^2 - \left(\sum_k \alpha_k \right)^2 \right). \end{aligned}$$

Setting $\alpha_k = g(\lambda_k)$ and $\alpha_k = \lambda_k$ and taking the difference,

$$\mathbb{E}[\text{Var}(s(\mathbf{Q})) - \text{Var}(s(\mathbf{M}))] = \frac{2}{n^2(n+2)} \left[n \sum_k g(\lambda_k)^2 - \left(\sum_k g(\lambda_k) \right)^2 - n \sum_k \lambda_k^2 + \left(\sum_k \lambda_k \right)^2 \right]. \quad (11)$$

Step 3: Reformulation via pairwise differences.

Applying the identity $n \sum_k x_k^2 - (\sum_k x_k)^2 = \sum_{i < j} (x_i - x_j)^2$ separately to $x_k = g(\lambda_k)$ and $x_k = \lambda_k$, we rewrite (11) as

$$\mathbb{E}[\text{Var}(s(\mathbf{Q})) - \text{Var}(s(\mathbf{M}))] = \frac{2}{n^2(n+2)} \sum_{1 \leq i < j \leq n} [(g(\lambda_i) - g(\lambda_j))^2 - (\lambda_i - \lambda_j)^2]. \quad (12)$$

Step 4: Applying the mean value theorem.

For each pair with $\lambda_i \neq \lambda_j$, the mean value theorem gives a point $\xi_{ij} \in (\min\{\lambda_i, \lambda_j\}, \max\{\lambda_i, \lambda_j\})$ such that

$$g(\lambda_i) - g(\lambda_j) = g'(\xi_{ij}) (\lambda_i - \lambda_j).$$

Squaring and substituting into (12),

$$(g(\lambda_i) - g(\lambda_j))^2 - (\lambda_i - \lambda_j)^2 = (g'(\xi_{ij})^2 - 1)(\lambda_i - \lambda_j)^2.$$

When $\lambda_i = \lambda_j$ both sides vanish trivially. Therefore

$$\mathbb{E}[\text{Var}(s(\mathbf{Q})) - \text{Var}(s(\mathbf{M}))] = \frac{2}{n^2(n+2)} \sum_{1 \leq i < j \leq n} (\lambda_i - \lambda_j)^2 (g'(\xi_{ij})^2 - 1),$$

which completes the proof. $\square$

B.2 Proof of Corollary 2.2

Proof. We divide the proof into three steps.

Step 1: $a > 1$ for Newton-Schulz iteration in Muon

In Muon, the input matrix is first Frobenius-normalized. Hence its singular values satisfy

$$0 \leq \sigma_k \leq 1, \quad k = 1, \dots, n,$$

and therefore

$$0 \leq \lambda_k = \sigma_k^2 \leq 1.$$

Thus the relevant regime for the Newton-Schulz map is the interval $[0, 1]$. Moreover, in Muon the Newton-Schulz step is designed to move singular values toward the target value 1, namely,

$$\varphi(x) > x \quad \text{for all } x \in (0, 1). \quad (13)$$

Since

$$\varphi(x) = x(a + bx^2 + cx^4) = ax + bx^3 + cx^5,$$

we have

$$\varphi(x) - x = (a - 1)x + bx^3 + cx^5 = x((a - 1) + bx^2 + cx^4).$$

Assume for contradiction that $a \leq 1$.

If $a < 1$, then

$$(a - 1) + bx^2 + cx^4 < 0$$

for all sufficiently small $x > 0$, because the constant term $a - 1$ is strictly negative and the higher-order terms vanish as $x \rightarrow 0$. Hence

$$\varphi(x) - x < 0$$

for all sufficiently small $x > 0$, contradicting (13). Without appealing to any specific choice of Newton-Schulz coefficients, we assume:

$$\varphi'(0) = a > 1.$$

Step 2: $a > 1$ implies that $|g'(\lambda)| > 1$ on a neighborhood of 0.

By definition,

$$g(\lambda) = \lambda(a + b\lambda + c\lambda^2)^2.$$

Differentiating gives

$$g'(\lambda) = (a + b\lambda + c\lambda^2)^2 + 2\lambda(a + b\lambda + c\lambda^2)(b + 2c\lambda).$$

In particular,

$$g'(0) = a^2.$$

Since $a > 1$, we get

$$g'(0) = a^2 > 1.$$

Because g' is continuous, there exists $\delta > 0$ such that

$$g'(\lambda)^2 > 1 \quad \text{for all } \lambda \in [0, \delta].$$

Step 3: Positivity of the variance shift.

By Lemma 2.1, for each pair $1 \leq i < j \leq n$, there exists

$$\xi_{ij} \in (\min\{\lambda_i, \lambda_j\}, \max\{\lambda_i, \lambda_j\})$$

such that

$$\mathbb{E}[\text{Var}(s(\mathbf{Q})) - \text{Var}(s(\mathbf{M}))] = \frac{2}{n^2(n+2)} \sum_{1 \leq i < j \leq n} (\lambda_i - \lambda_j)^2 (g'(\xi_{ij})^2 - 1).$$

Now suppose that

$$\lambda_1, \dots, \lambda_n \in [0, \delta].$$

Then for every $i < j$,

$$\xi_{ij} \in [0, \delta],$$

and therefore

$$g'(\xi_{ij})^2 - 1 > 0.$$

Hence each summand

$$(\lambda_i - \lambda_j)^2 (g'(\xi_{ij})^2 - 1)$$

is nonnegative. Since the λ_k are not all equal, there exists at least one pair $i < j$ such that

$$(\lambda_i - \lambda_j)^2 > 0,$$

and for that pair the corresponding summand is strictly positive. Therefore the whole sum is strictly positive, and since

$$\frac{2}{n^2(n+2)} > 0,$$

we conclude that

$$\mathbb{E}[\text{Var}(s(\mathbf{Q})) - \text{Var}(s(\mathbf{M}))] > 0.$$

This proves the claim. □

C Pseudocode of Muon+

Algorithm 1 Python code for the MUON+ update.

```
1 def muon_plus_step(W, M_prev, G, mu, lr, d="col", eps=1e-8):
2     # momentum
3     M = mu * M_prev + (1.0 - mu) * G
4     # orthogonalize
5     U = Ortho(M) # newton-schulz
6     # normalize
7     O = norm_dir(U, d=d, eps=eps)
8     # update
9     m, n = W.shape[-2], W.shape[-1]
10    W = W - lr * (m / n) ** 0.5 * O
11    return W, M
12 def norm_dir(X, d="col", eps=1e-8):
13     if d == "col":
14         denom = (X.square().sum(dim=-2, keepdim=True) + eps).sqrt()
15         return X / denom
16     if d == "row":
17         denom = (X.square().sum(dim=-1, keepdim=True) + eps).sqrt()
18         return X / denom
19     if d == "col_row":
20         return norm_dir(norm_dir(X, "col", eps), "row", eps)
21     if d == "row_col":
22         return norm_dir(norm_dir(X, "row", eps), "col", eps)
```

D Regions of Imbalance Amplification and Reduction

D.1 Jordan Coefficients [15]

In [15], the coefficients are set to

$$(a, b, c) = (3.4445, -4.7750, 2.0315).$$

The resulting regions are as Figure 8:

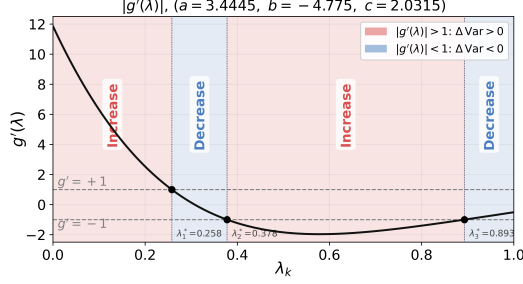


Figure 8: Regions of imbalance amplification and reduction for Jordan Coefficients.

D.2 You Coefficients [6]

You [6] further optimizes the coefficients at each iteration:

$$\begin{aligned} \text{coeff} = & [(3955/1024, -8306/1024, 5008/1024), \\ & (3735/1024, -6681/1024, 3463/1024), \\ & (3799/1024, -6499/1024, 3211/1024), \\ & (4019/1024, -6385/1024, 2906/1024), \\ & (2677/1024, -3029/1024, 1162/1024), \\ & (2172/1024, -1833/1024, 682/1024)]. \end{aligned}$$

Then the regions are as follows per step:

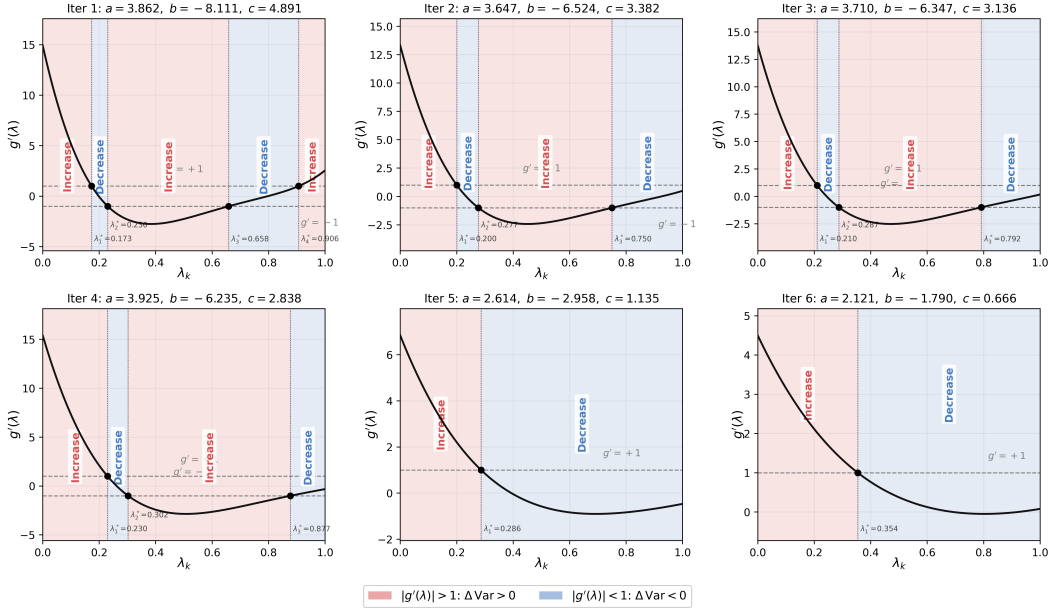


Figure 9: Regions of imbalance amplification and reduction for You Coefficients.

D.3 PolarExpress Coefficients [2]

[2] uses iteration-dependent coefficients:

$$\{(a_t, b_t, c_t)\}_{t=1}^8 = [(8.28721201814563, -23.595886519098837, 17.300387312530933), \\ (4.107059111542203, -2.9478499167379106, 0.5448431082926601), \\ (3.9486908534822946, -2.908902115962949, 0.5518191394370137), \\ (3.3184196573706015, -2.488488024314874, 0.51004894012372), \\ (2.300652019954817, -1.6689039845747493, 0.4188073119525673), \\ (1.891301407787398, -1.2679958271945868, 0.37680408948524835), \\ (1.8750014808534479, -1.2500016453999487, 0.3750001645474248), \\ (1.875, -1.25, 0.375)],$$

with subsequent coefficients numerically equal to $(1.875, -1.25, 0.375)$.

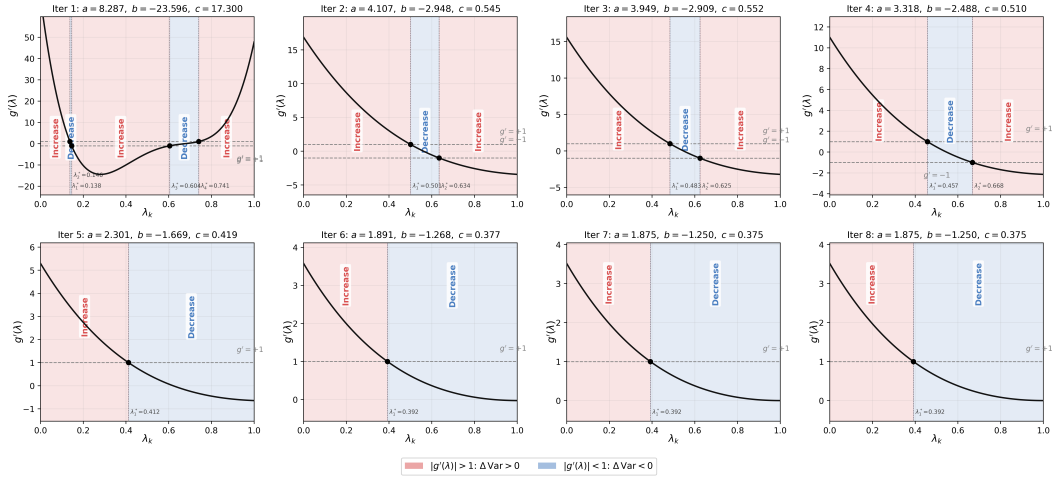


Figure 10: Regions of imbalance amplification and reduction for PolarExpress Coefficients.

E Hyperparameter

E.1 Model Configurations

Model	n_{embd}	n_{layer}	n_{head}	Param(M)
GPT-Small	768	12	12	124
GPT-Base	1024	24	16	362
GPT-Large	1280	36	20	774
GPT-Huge	4096	32	32	6654

Table 10: Architecture configurations of GPT models.

Model	n_{embd}	n_{layer}	n_{head}	FFN dim	Param(M)
60M	512	8	8	1376	58
130M	768	12	12	2048	134
350M	1024	24	16	2736	368
1B	2048	24	32	5461	1339
7B	4096	32	32	11008	6738

Table 11: Architecture configurations of LLaMA-style models.

E.2 Training Configurations

E.2.1 GPT Models

Model	Hyperparameter	Muon	MUON+
GPT-Small	Sequence length	2048	2048
	Learning rate	0.005	0.01
	Weight decay	0.1	0.1
	Norm _d	—	Col-Row
GPT-Base	Sequence length	4096	4096
	Learning rate	0.005	0.005
	Weight decay	0.1	0.1
	Norm _d	—	Row
GPT-Large	Sequence length	8192	8192
	Learning rate	0.02	0.01
	Weight decay	0.1	0.1
	Norm _d	—	Row-Col
GPT-Huge	Sequence length	1024	1024
	Learning rate	0.01	0.01
	Weight decay	0.005	0.005
	Norm _d	—	Row-Col

Table 12: Best training hyperparameters for GPT-Small/Base/Large/Huge on FineWeb. Sequence lengths are set to 2048/4096/8192 for Small/Base/Large, respectively. We use Jordan orthogonalization for all runs. We keep the same learning rate scheduler as in NanoGPT: a constant learning rate for the first 40% of training steps followed by a linear decay to zero. Sweeping results are provided in Appendix F.

Model	Hyperparameter	Muon	MUON+
GPT-Base	Learning rate	0.005	0.005
	Weight decay	0.1	0.1
	Norm _d	—	Row-Col

Table 13: Training hyperparameters for overtraining GPT-Base. We train 72 billion FineWeb tokens for each setting. Sequence length is 4096 for both runs.

E.2.2 LLaMA Models

Model	Hyperparameter	Muon	MUON+
LLaMA-60M	Sequence length	1024	1024
	Learning rate	0.06	0.06
	LR scheduler	Cosine	Cosine
	Weight decay	0.1	0.1
	Warmup ratio	0.1	0.1
	Norm _d	—	Row-Col
LLaMA-130M	Sequence length	1024	1024
	Learning rate	0.02	0.02
	LR scheduler	Cosine	Cosine
	Weight decay	0.1	0.1
	Warmup ratio	0.1	0.1
	Norm _d	—	Col-Row
LLaMA-350M	Sequence length	4096	4096
	Learning rate	0.04	0.04
	LR scheduler	Cosine	Cosine
	Weight decay	0.1	0.1
	Warmup ratio	0.1	0.1
	Norm _d	—	Col-Row
LLaMA-1B	Sequence length	4096	4096
	Learning rate	0.02	0.02
	LR scheduler	Cosine	Cosine
	Weight decay	0.1	0.1
	Warmup ratio	0.1	0.1
	Norm _d	—	Row-Col
LLaMA-7B	Sequence length	1024	1024
	Learning rate	0.01	0.01
	LR scheduler	Cosine	Cosine
	Weight decay	0.005	0.005
	Warmup ratio	0.1	0.1
	Norm _d	—	Row-Col

Table 14: Best training hyperparameters for LLaMA on FineWeb. Sequence lengths are set to 1024 for 60M/130M and 4096 for 350M/1B. All runs use Jordan orthogonalization. Norm_(col_row) and Norm_(row_col) yield nearly identical performance. Additional sweeping results are provided in Appendix F.

Model	Hyperparameter	Muon	MUON+
LLaMA-350M	Learning rate	0.04	0.04
	Weight decay	0.1	0.1
	Norm _d	—	Col-Row

Table 15: Training hyperparameters for overtraining LLaMA-350M. We train 72 billion FineWeb tokens for each setting. Sequence length is 4096 for both runs.

F Detailed Sweep Experiments

LR	None	Col	Row	Col-Row	Row-Col
<i>LLaMA-60M</i>					
0.005	28.98	28.17	28.14	28.10	28.03
0.01	27.03	26.59	26.42	26.46	26.41
0.02	26.22	25.83	25.69	25.65	25.68
0.04	25.90	25.49	25.30	25.28	25.28
0.06	25.75	25.34	25.29	25.25	25.25
0.08	25.97	25.39	25.41	25.29	25.28
<i>LLaMA-130M</i>					
0.005	20.40	20.05	19.89	19.86	19.87
0.01	19.34	19.16	18.98	18.98	19.01
0.02	19.06	18.85	18.67	18.65	18.68
0.04	19.35	19.00	18.92	18.87	18.87
0.06	19.70	19.31	19.25	19.21	19.20
0.08	20.14	19.63	19.61	19.56	19.54
<i>LLaMA-350M</i>					
0.005	15.54	14.97	14.94	14.92	14.86
0.01	14.48	14.18	14.00	13.96	13.95
0.02	14.11	13.73	13.46	13.43	13.46
0.04	14.02	13.61	13.44	13.41	13.44
0.06	14.23	13.65	13.48	13.50	13.46
0.08	14.56	13.76	13.60	13.60	13.61
<i>LLaMA-1B</i>					
0.005	11.47	-	-	10.95	10.92
0.01	10.88	-	-	10.50	10.48
0.02	10.68	-	-	10.32	10.31
0.04	10.74	-	-	10.42	10.43
0.06	10.91	-	-	10.57	10.61
0.08	11.15	-	-	10.77	10.78

Table 16: Best validation perplexity per norm setting and learning rate for all LLaMA models. Bold marks the best entry in each row. **None** denotes the Muon baseline.

LR	None	Col	Row	Col-Row	Row-Col
<i>GPT-Small</i>					
0.003	31.10	29.75	28.37	28.39	29.01
0.005	29.66	28.81	27.97	27.91	28.09
0.01	29.87	28.08	27.69	27.64	27.76
0.02	30.13	29.03	29.27	28.56	27.72
0.04	30.14	29.71	30.25	30.58	29.37
<i>GPT-Base</i>					
0.003	22.10	21.37	20.22	20.20	20.75
0.005	21.70	20.63	19.98	20.03	20.11
0.01	21.89	20.73	20.93	20.74	20.35
0.02	21.75	20.83	21.15	21.10	20.27
0.04	22.14	23.27	23.16	23.41	22.12
<i>GPT-Large</i>					
0.005	18.26	-	-	-	17.16
0.01	18.28	-	-	-	16.91
0.02	17.82	-	-	-	17.52
0.04	18.84	-	-	-	17.83

Table 17: Best validation perplexity per norm setting and learning rate for GPT Models. Bold marks the best entry in each row. **None** is the Muon baseline.

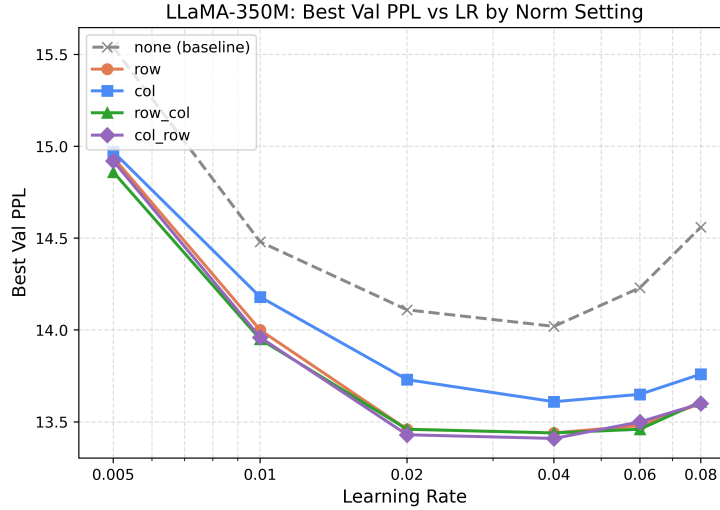


Figure 11: Visualization of sweep result for LLaMA-350M

Model	Muon	MUON+
GPT-Base	21.68 ± 0.050	19.98 ± 0.031
LLaMA-350M	14.06 ± 0.046	13.42 ± 0.020

Table 18: Mean validation perplexity over 5 random seeds. We report mean $\pm$ standard deviation for GPT-Base and LLaMA-350M. Lower is better. Hyperparameters are consistent with Table 17 and 16.

Remark. The proposed normalization changes the Frobenius norm of the update matrix and may thus induce an implicit layer-wise learning-rate rescaling. One may worry that the gains of MUON+

come from this effect rather than from reduced parameter-space variance. However, our learning-rate sweeps show that the optimal learning rate of MUON+ is essentially unchanged from that of Muon (Tables 16 and 17, Figure 11), suggesting that the improvement is not due to a shift in the effective learning rate. We further verify this by matching the Frobenius norm of the normalized update to its pre-normalization value. The results are shown in Table 19.

Model	Muon	MUON+
GPT-Base	21.70	20.16
LLaMA-350M	14.02	13.55

Table 19: Matching Frobenius norm. Hyperparameters are consistent with Table 17 and 16.

G Supplementary results

G.1 Per-step runtime

The results are measured on an H100 GPU with mixed precision (bf16), batch size 4, and sequence length 4096, and are reported in ms/step. Compared to Muon, all normalization variants introduce only negligible runtime overhead.

Method	Time / step	Ratio
Muon	892.8	1.000
MUON+ (col)	896.6	1.004
MUON+ (row)	900.6	1.008
MUON+ (col_row)	909.9	1.019
MUON+ (row_col)	910.3	1.019

Table 20: Per-step runtime comparison between Muon and MUON+.

G.2 Imbalance trend

